

EKG Reading Notes: A Comprehensive Reference

Table of Contents

1. Foundations
 2. Waveform Components
 3. A Systematic Approach
 4. Rate
 5. Rhythm Analysis
 6. Axis Determination
 7. Intervals and Segments
 8. Conduction Abnormalities
 9. Chamber Enlargement and Hypertrophy
 10. Ischemia and Infarction
 11. Electrolyte Abnormalities
 12. Important "Can't-Miss" Patterns
 13. Pacemakers
 14. Artifact vs. Pathology
 15. Quick Reference Summary
-

1. Foundations

What an EKG Measures

An electrocardiogram records the electrical activity of the heart from the body's surface. Each lead views the heart from a different angle, producing a unique tracing. The size and direction of each deflection reflect the net vector of electrical depolarization or repolarization relative to that lead's axis.

The 12-Lead System

Limb leads (frontal plane):

- **Bipolar leads:** I, II, III
- **Augmented leads:** aVR, aVL, aVF

Precordial leads (horizontal plane): V1 through V6

Lead groupings by anatomy:

Region	Leads	Artery (usually)
Inferior	II, III, aVF	RCA (80%) or LCx
Lateral	I, aVL, V5, V6	LCx or diagonal
Anterior	V3, V4	LAD
Septal	V1, V2	LAD (proximal)
High lateral	I, aVL	LCx
Posterior	Mirror in V1-V3	RCA or LCx

Paper Speed and Calibration

- **Standard speed:** 25 mm/second
- **Standard calibration:** 10 mm = 1 mV
- **Small box:** 1 mm wide (0.04 sec), 1 mm tall (0.1 mV)
- **Large box:** 5 mm wide (0.20 sec), 5 mm tall (0.5 mV)
- Always confirm the calibration square wave before interpreting — half-standard or double-standard tracings will mislead you.

2. Waveform Components

P Wave

Represents atrial depolarization.

- **Normal duration:** < 0.12 sec (< 3 small boxes)
- **Normal amplitude:** < 2.5 mm
- **Best seen in:** Lead II (for rhythm) and V1 (for morphology)
- Upright in I, II, aVF; inverted in aVR (in sinus rhythm)

PR Interval

From start of P to start of QRS — atrial depolarization plus AV node delay.

- **Normal:** 0.12–0.20 sec (3–5 small boxes)

- Short → pre-excitation (e.g., WPW)
- Long → first-degree AV block

QRS Complex

Ventricular depolarization.

- **Normal duration:** < 0.12 sec (< 3 small boxes); narrow complexes are supraventricular in origin, wide complexes suggest ventricular origin or aberrant conduction.
- **Q wave:** first negative deflection
- **R wave:** first positive deflection
- **S wave:** negative deflection following R

ST Segment

End of QRS (J point) to start of T wave. Should be isoelectric. Elevation or depression can indicate ischemia, injury, pericarditis, early repolarization, or LV strain.

T Wave

Ventricular repolarization.

- Usually upright in I, II, V3-V6
- Inverted in aVR
- Variable in III, aVL, V1-V2
- Tall, peaked T waves → hyperkalemia or hyperacute ischemia
- Inverted T waves → ischemia, strain, CNS events

QT Interval

From start of QRS to end of T. Reflects total ventricular depolarization plus repolarization. Must be **corrected for rate (QTc)** using Bazett's formula: $QTc = QT / \sqrt{RR}$.

- **Normal QTc:** < 440 ms (men), < 460 ms (women)
- **Prolonged QTc > 500 ms:** significant risk of torsades de pointes

U Wave

Small deflection after T. Prominent U waves suggest hypokalemia, bradycardia, or drug effect.

3. A Systematic Approach to Reading Every EKG

Never interpret piecemeal. Use the same sequence every time so nothing gets missed:

1. **Confirm patient data and calibration** (name, date, standard or half-standard)
 2. **Rate**
 3. **Rhythm**
 4. **Axis**
 5. **Intervals** (PR, QRS, QT/QTc)
 6. **P wave morphology**
 7. **QRS morphology** (voltage, Q waves, R-wave progression)
 8. **ST segment** (elevation, depression, J-point)
 9. **T waves** (direction, symmetry, amplitude)
 10. **Overall clinical impression** — always synthesize findings with the patient in front of you
-

4. Rate

Methods

- **300 method:** Count large boxes between two R waves; divide 300 by that number.
Memorize: 300, 150, 100, 75, 60, 50, 43, 37.
- **1500 method:** Count small boxes between R waves; divide 1500 by that number. Most precise for regular rhythms.
- **6-second method:** Count QRS complexes in 30 large boxes (6 sec) and multiply by 10. Best for irregular rhythms.

Definitions

- **Bradycardia:** < 60 bpm
 - **Normal:** 60-100 bpm
 - **Tachycardia:** > 100 bpm
-

5. Rhythm Analysis

Criteria for Normal Sinus Rhythm

- Rate 60-100 bpm
- Regular R-R intervals

- P before every QRS, QRS after every P
- P waves upright in I, II, aVF; inverted in aVR
- Constant PR interval (0.12–0.20 sec)

Sinus Variants

- **Sinus bradycardia:** NSR with rate < 60
- **Sinus tachycardia:** NSR with rate > 100
- **Sinus arrhythmia:** phasic variation with respiration (common and benign in young patients)

Atrial Arrhythmias

Premature Atrial Contraction (PAC): Early P wave with abnormal morphology, usually followed by a narrow QRS.

Atrial Fibrillation:

- Irregularly irregular rhythm
- No discernible P waves — chaotic fibrillatory baseline
- Variable ventricular response

Atrial Flutter:

- "Sawtooth" flutter waves, typically at ~300 bpm atrial rate
- Commonly conducts 2:1 (ventricular rate ~150 bpm)
- Regular unless variable block present

Multifocal Atrial Tachycardia (MAT):

- Rate > 100
- At least 3 distinct P-wave morphologies
- Often in COPD patients

Supraventricular Tachycardia (SVT):

- Narrow complex, regular, rate typically 150–250
- P waves often buried in QRS or appear as retrograde pseudo-R' in V1

Junctional Rhythms

- Narrow QRS, rate 40–60 (junctional escape) or 60–100 (accelerated)
- Inverted or absent P waves, or P buried in QRS

Ventricular Arrhythmias

Premature Ventricular Contraction (PVC): Wide, bizarre QRS without preceding P; typically followed by a compensatory pause.

Ventricular Tachycardia (VT):

- ≥ 3 consecutive PVCs at rate > 100
- Wide QRS (> 0.12 sec)
- Monomorphic or polymorphic
- **Sustained** if > 30 seconds or hemodynamically unstable
- Treat any wide-complex tachycardia as VT until proven otherwise

Torsades de Pointes: Polymorphic VT with QRS amplitude that twists around the baseline; arises in the setting of prolonged QT.

Ventricular Fibrillation: Chaotic, irregular deflections without identifiable QRS complexes. Immediately fatal without defibrillation.

Asystole: Flat line (confirm in more than one lead to exclude lead disconnection).

Idioventricular rhythm: Wide QRS, rate 20–40, no P waves. "Accelerated" idioventricular rhythm runs 40–120 and is a classic reperfusion rhythm post-MI.

6. Axis Determination

The "axis" refers to the net direction of ventricular depolarization in the frontal plane.

Quick Method — The Two-Lead Shortcut

Lead I	Lead aVF	Axis
Positive	Positive	Normal (-30° to $+90^\circ$)
Positive	Negative	Possible Left Axis Deviation (check II)
Negative	Positive	Right Axis Deviation

Lead I	Lead aVF	Axis
Negative	Negative	Extreme axis ("Northwest")

If Lead I is positive and aVF is negative, look at Lead II: if II is also positive, the axis is still normal; if II is negative, it's LAD.

Common Causes

- **Left axis deviation:** LVH, left anterior fascicular block, inferior MI, WPW
- **Right axis deviation:** RVH, left posterior fascicular block, lateral MI, PE, pediatric normal
- **Extreme axis:** Ventricular rhythm, severe RVH, hyperkalemia

7. Intervals and Segments

PR Interval

- **Short (< 120 ms):** WPW, LGL, junctional rhythm
- **Long (> 200 ms):** First-degree AV block

QRS Duration

- **Narrow (< 120 ms):** Supraventricular origin
- **Wide (≥ 120 ms):** Bundle branch block, ventricular rhythm, hyperkalemia, sodium channel blocker toxicity, pre-excitation

QT / QTc

- Measure in lead II or V5, ideally where T waves are most distinct
- QTc > 500 ms raises torsades risk significantly
- Common causes of prolonged QT: congenital long QT syndromes, hypokalemia, hypomagnesemia, hypocalcemia, antiarrhythmics (sotalol, amiodarone), antipsychotics, macrolides, fluoroquinolones, methadone

8. Conduction Abnormalities

AV Blocks

First-degree AV block:

- PR > 200 ms, constant
- Every P conducts; benign in isolation

Second-degree AV block, Mobitz I (Wenckebach):

- Progressive PR prolongation until a P fails to conduct
- R-R interval shortens before the dropped beat
- Usually AV nodal, often benign, rarely requires pacing

Second-degree AV block, Mobitz II:

- Fixed PR interval with intermittent dropped QRS
- Typically infranodal (His-Purkinje)
- High risk of progression to complete block; usually requires pacing

Third-degree (complete) AV block:

- Complete AV dissociation
- Atrial and ventricular rates independent
- Escape rhythm determines QRS width: narrow (junctional escape) or wide (ventricular escape)
- Requires pacing

Bundle Branch Blocks

Right Bundle Branch Block (RBBB):

- QRS $\geq$ 120 ms
- rSR' ("bunny ears") in V1
- Wide slurred S wave in I, V6
- Normal axis usually preserved

Left Bundle Branch Block (LBBB):

- QRS $\geq$ 120 ms

- Broad monomorphic R in I, V5, V6 (often notched)
- Deep S or QS pattern in V1
- ST and T wave changes are expected (discordant to QRS); **new LBBB in the setting of chest pain is a STEMI equivalent** — consider Sgarbossa criteria for ischemia in LBBB

Fascicular Blocks (Hemiblocks)

Left Anterior Fascicular Block (LAFB):

- Left axis deviation (-45° to -90°)
- Small Q in I and aVL, small R in II, III, aVF
- QRS < 120 ms

Left Posterior Fascicular Block (LPFB):

- Right axis deviation ($+90^\circ$ to $+180^\circ$)
- Small R in I and aVL, small Q in II, III, aVF
- QRS < 120 ms
- Much rarer; diagnose only after excluding RVH, lateral MI

Bifascicular block: RBBB + LAFB or RBBB + LPFB. **Trifascicular block:** Bifascicular block + first-degree AV block (historical term; imprecise but still used).

9. Chamber Enlargement and Hypertrophy

Left Atrial Enlargement (LAE)

- Lead II: P wave > 120 ms, notched ("P mitrale")
- V1: terminal negative deflection ≥ 1 mm deep and ≥ 40 ms wide

Right Atrial Enlargement (RAE)

- Lead II: P wave ≥ 2.5 mm tall ("P pulmonale")
- V1: initial positive deflection ≥ 1.5 mm

Left Ventricular Hypertrophy (LVH)

Sokolow-Lyon: S in V1 + R in V5 or V6 ≥ 35 mm **Cornell voltage:** R in aVL + S in V3 > 28 mm (men) or > 20 mm (women) **Romhilt-Estes point score:** more specific, uses multiple criteria

Associated findings: LAE, left axis deviation, repolarization abnormalities (LV "strain" — downsloping ST depression and T-wave inversion in lateral leads).

Right Ventricular Hypertrophy (RVH)

- Right axis deviation
- R/S ratio > 1 in V1
- Deep S waves in V5, V6
- Right atrial enlargement often coexists
- RV "strain": T-wave inversion in V1-V3

10. Ischemia and Infarction

Recognizing STEMI

Criteria (new guidelines): ST elevation at the J point in two contiguous leads:

- ≥ 1 mm in most leads
- ≥ 2 mm (men) or ≥ 1.5 mm (women) in V2-V3
- ≥ 0.5 mm in V7-V9 (posterior) or V3R-V4R (right-sided)

Localization of Infarction

Location	ST Elevation	Reciprocal ST Depression	Likely Artery
Anterior	V1-V4	II, III, aVF	LAD
Anteroseptal	V1-V2	—	LAD (proximal)
Lateral	I, aVL, V5, V6	II, III, aVF	LCx or diagonal
Inferior	II, III, aVF	I, aVL	RCA (or LCx)
Posterior	V7-V9 (reciprocal ST depression in V1-V3 with tall R)	V1-V3	RCA or LCx
Right ventricular	V3R, V4R	—	Proximal RCA

Always Get Right-Sided and Posterior Leads for Inferior MI

Up to 40% of inferior MIs have RV involvement. Missing this is clinically important: nitrates can cause severe hypotension in RV infarct; treatment is fluids.

Non-STEMI / Ischemic Patterns

- **ST depression** (horizontal or downsloping, ≥ 0.5 mm): subendocardial ischemia
- **T-wave inversion** (symmetric, deep): ischemia or post-ischemic
- **Dynamic changes** on serial EKGs carry more weight than a single tracing

Q Waves

Pathologic Q waves: ≥ 40 ms wide OR $> 25\%$ of R-wave height in that lead. Suggest completed (old or evolving) infarction. Exception: Q in aVR and isolated small Q in V1 can be normal.

High-Risk Patterns You Must Recognize

Wellens Syndrome: Deep symmetric T-wave inversions (Type B) or biphasic T waves (Type A) in V2-V3, typically in a pain-free patient. Indicates a critical proximal LAD stenosis. Needs cath, not stress testing.

De Winter T Waves: Upsloping ST depression at the J point in precordial leads with tall, symmetric T waves. STEMI equivalent — proximal LAD occlusion.

Sgarbossa Criteria (for MI in LBBB or paced rhythm):

- Concordant ST elevation ≥ 1 mm: 5 points
- Concordant ST depression ≥ 1 mm in V1-V3: 3 points
- Discordant ST elevation ≥ 5 mm: 2 points
- Score ≥ 3 is specific for MI. The **modified Smith-Sgarbossa** replaces the last criterion with a ratio of ST elevation to S-wave depth ≤ -0.25 and is more sensitive.

aVR ST elevation > 1 mm with diffuse ST depression suggests left main disease, triple-vessel disease, or diffuse subendocardial ischemia.

11. Electrolyte Abnormalities

Potassium

Hyperkalemia (progressive changes):

1. Peaked, narrow-based T waves (early; K ~ 5.5 - 6.5)

2. P wave flattening and PR prolongation
3. QRS widening
4. Merging of QRS with T → "sine wave" (imminent arrest)
5. Asystole or VF

Hypokalemia:

- Flattened T waves
- ST depression
- Prominent U waves
- Prolonged QT (actually QU)
- Predisposes to torsades

Calcium

- **Hypercalcemia:** Short QT, short ST segment, Osborn J waves in severe cases
- **Hypocalcemia:** Prolonged QT (prolonged ST segment specifically)

Magnesium

- **Hypomagnesemia:** Often mirrors hypokalemia; promotes torsades
 - **Hypermagnesemia:** Bradycardia, PR/QRS prolongation, AV block
-

12. Important "Can't-Miss" Patterns

Wolff-Parkinson-White (WPW)

- Short PR (< 120 ms)
- Delta wave (slurred upstroke of QRS)
- Wide QRS
- Secondary ST-T changes
- Risk: rapid atrial fibrillation with conduction down the accessory pathway → VF. Avoid AV-nodal blockers (adenosine, beta blockers, verapamil, digoxin) in AF with WPW.

Brugada Syndrome

- Type 1: Coved ST elevation ≥ 2 mm in V1-V2 with T-wave inversion

- Type 2: Saddleback ST elevation
- Associated with sudden cardiac death; genetic sodium channel disorder (SCN5A)

Long QT Syndromes

Congenital or acquired. QTc > 500 ms is the danger zone. Seek and correct electrolytes and QT-prolonging medications.

Pericarditis

- **Diffuse concave-up** ST elevation (not confined to a vascular territory)
- **PR depression** (especially in II; PR elevation in aVR)
- Typically no reciprocal ST depression (except aVR/V1)
- Evolves through four Spodick stages

Pulmonary Embolism

- Sinus tachycardia is the most common finding
- **S1Q3T3** pattern (deep S in I, Q and inverted T in III) — classic but uncommon
- Right axis deviation, new incomplete RBBB
- T-wave inversions in V1-V4 (right heart strain)

Digoxin Effect vs. Toxicity

- **Effect:** "Scooped" / "Salvador Dalí mustache" ST depression, short QT
- **Toxicity:** Virtually any arrhythmia, but classic findings include atrial tachycardia with block, bidirectional VT, frequent PVCs

Hypothermia

- **Osborn (J) waves** — positive deflection at J point
- Bradycardia, prolonged intervals, muscle tremor artifact

Early Repolarization

- Notched or slurred J point with concave ST elevation
- Usually in young, healthy patients
- Most common in precordial leads
- Diagnosis of exclusion

Takotsubo ("Stress") Cardiomyopathy

- Often mimics anterior STEMI
 - Deep T-wave inversions and marked QT prolongation in evolving phase
 - Clinical context matters: emotional stressor, post-menopausal woman, apical ballooning on echo, clean coronaries on angiography
-

13. Pacemakers

Pacing Spikes

Sharp vertical deflections immediately preceding the paced complex. Modern bipolar pacers produce small spikes that can be subtle.

Nomenclature (NASPE/BPEG Code)

Five letters; first three are most used clinically:

1. Chamber paced (A, V, D = both, O = none)
2. Chamber sensed
3. Response to sensing (I = inhibited, T = triggered, D = both)

Examples:

- **VVI**: Ventricular pace, ventricular sense, inhibited by native beats
- **DDD**: Dual-chamber pace and sense with both inhibition and triggering

Troubleshooting

- **Failure to capture**: Spike without following depolarization (lead dislodgement, battery depletion, electrolytes)
- **Failure to sense (undersensing)**: Pacer fires despite native beats
- **Oversensing**: Pacer inhibited when it shouldn't be (T-wave oversensing, myopotentials)

Ischemia in a Paced Rhythm

Use modified Sgarbossa criteria, as with LBBB.

Common artifacts that fool readers:

- **Lead reversal** (LA/RA): inverted P and QRS in lead I — mimics dextrocardia
- **Muscle tremor**: rapid irregular baseline (e.g., Parkinson's, shivering) — can mimic AF
- **60-Hz interference**: regular fine oscillation
- **Wandering baseline**: respiration or loose electrode
- **Lead disconnection**: flat line in a single lead

Always confirm dysrhythmias in more than one lead before acting.

15. Quick Reference Summary

Normal Values at a Glance

Parameter	Normal Range
Rate	60–100 bpm
P wave	< 120 ms, < 2.5 mm
PR interval	120–200 ms
QRS	< 120 ms
QTc	< 440 ms (M) / < 460 ms (W)
R-wave progression	R > S by V3–V4
Axis	–30° to +90°

Always Ask Yourself

1. Is this rhythm stable?
2. Is this a STEMI or STEMI equivalent?
3. Is the QT dangerously long?
4. Is the QRS unexpectedly wide — and why?
5. Does the EKG match the clinical picture?

The Most Common Traps

- Assuming a wide-complex tachycardia is SVT with aberrancy — it's VT until proven otherwise
- Missing posterior or right-sided MI because you only looked at the standard 12 leads
- Calling it sinus rhythm when the "P" is actually a retrograde or flutter wave
- Reading a half-standard or double-standard EKG at face value
- Treating the tracing without looking at the patient

These notes are a study and clinical reference aid. Always correlate EKG findings with clinical context, serial tracings, and appropriate advanced imaging or biomarker data. Do not use any document as a substitute for formal training, supervised practice, or specialist consultation in real patient care.